

Lecture 5.2: Data Collection via APIs

AREC 570 / ECON 530 / Unit 5

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Reading and sequence

Dominitz and Manski, “More Data or Better Data? A Statistical Decision Problem” ([Dominitz and Manski 2017](#))

1. Connect APIs to secondary data collection
2. Retrieve four types of data in R
3. Check what each downloaded observation represents
4. Document a reproducible retrieval
5. Share the current data plan for each research project

Readings and course handouts are available on [Canvas](#)

APIs deliver secondary data

The organization responsible for the source determines

- What is measured
- Which people, places, or activities are covered
- When observations are collected and released
- How uncertainty, missing values, and revisions are recorded

An **application programming interface** provides a documented way for R to request selected records from that source

What happens when R calls an API

R function → request → data service → response → R object

- **Endpoint** is the web address for a data service
- **Parameters** specify the place, time, variables, and other selections
- **Response** is the information returned by the service
- **R package** translates the response into a familiar object

Many services transmit a structured text format called **JSON**. The packages in today's examples read that format and return a data frame, map, or raster

Four applications

Application	Source	Returned to R
Land use	USDA Cropland Data Layer	Raster map or summary table
Water	USGS Water Data	Daily time series
Weather	NASA POWER	Daily weather table
Population	Census ACS	Estimates margins of error and map geometry

Each example begins with a small request that can be inspected before the request is expanded

Run R in the HTML slides

- Wait for the R ready message after opening the HTML lecture
- Select **Run Code** to execute a live cell
- Change a named value such as `threshold` or `selected`
- Select **Start Over** to restore the original code

The API retrieval slides show the complete request code for RStudio. The live cells use compact classroom extracts shaped like the returned records, so every student can run them from the course site

The PDF contains the initial code for reference

Install the R packages

Run this installation once on your computer

```
1 install.packages(c(  
2   "CropScapeR",  
3   "dataRetrieval",  
4   "nasapower",  
5   "tidycensus",  
6   "ggplot2",  
7   "dplyr"  
8 ))
```

Create a folder for files returned by the services

```
1 dir.create("data/raw", recursive = TRUE, showWarnings = FALSE)
```

A retrieval workflow

1. Read the source documentation
2. Define the place, time period, variables, and unit of observation
3. Run a small request
4. Inspect column names, values, units, and missingness
5. Save the returned file or object
6. Record the request and retrieval date
7. Expand the request after the checks are complete

Example 1: land use

Question

Which crop and land-cover classes does the Cropland Data Layer assign in Larimer County in 2024?

Source

USDA NASS Cropland Data Layer through the CropScape service

Observation

Each raster cell has a modeled crop or land-cover class for a particular year

[CropScape documentation](#) | [CropScapeR documentation](#)

Request a Cropland Data Layer map

```
1 library(CropScapeR)
2
3 larimer_cdl <- GetCDLData(
4   aoi = "08069",
5   year = 2024,
6   type = "f",
7   save_path = "data/raw/cdl_larimer_2024.tif"
8 )
9
10 raster::plot(larimer_cdl)
```

08069 is the five-digit Census code for Larimer County

The saved GeoTIFF contains a grid of land-cover codes and the information needed to locate those cells on a map

Use the land-use data

- Map crop locations
- Calculate acreage by crop class
- Compare annual crop patterns
- Define land-use exposure near water infrastructure or development

Try it: summarize land-cover records

Change `selected`, then run the cell. The classroom extract has the same class-and-acreage structure used in a CDL summary

R Code ↺ Start Over

▶ Run Code

```
1 cdl <- data.frame(  
2   class = c("Corn", "Alfalfa", "Grassland", "Developed", "Other"),  
3   acres = c(14200, 8900, 180500, 72400, 12600)  
4 )  
5 selected <- c("Corn", "Alfalfa")  
6 chosen <- subset(cdl, class %in% selected)  
7 chosen$county_share <- round(chosen$acres / sum(cdl$acres), 3)  
8 chosen
```

Assess the land-use data

- Crop-class definitions for each year
- Classification accuracy for the study region and crop
- Pixel resolution and coordinate system
- Treatment of double cropping and small fields
- Consistency of the study boundary across years

The CDL records modeled land-cover classes. Parcel ownership, zoning, and management practices require additional sources

Example 2: water

Question

What was the daily flow of the Cache la Poudre River during the 2024 growing season?

Source

USGS Water Data API

Observation

Each row reports a daily statistic for one parameter at one monitoring location

[USGS monitoring location 06752260](#) | [dataRetrieval tutorial](#)

Request daily streamflow

```
1 library(dataRetrieval)
2
3 poudre_flow <- read_waterdata_daily(
4   monitoring_location_id = "USGS-06752260",
5   parameter_code = "00060",
6   time = c("2024-04-01", "2024-09-30")
7 )
8
9 names(poudre_flow)
10 head(poudre_flow)
```

`00060` is the USGS parameter code for discharge

The returned columns include date, value, unit, approval status, qualifiers, station geometry, and a time-series identifier

Try it: define a low-flow threshold

Change `threshold`, then run the cell. The compact series represents the date and value columns returned by the water request

R Code [↺ Start Over](#)

[▶ Run Code](#)

```
1 poudre_flow <- data.frame(  
2   date = as.Date("2024-07-01") + 0:13,  
3   flow = c(970, 910, 820, 760, 680, 610, 540, 490, 450, 420, 390, 360, 335, 310)  
4 )  
5 threshold <- 500  
6 cat("Days below threshold:", sum(poudre_flow$flow < threshold))  
7 plot(flow ~ date, poudre_flow, type = "l", xlab = "", ylab = "Daily discharge")  
8 abline(h = threshold, col = "#C4513A", lty = 2)
```

Assess the streamflow series

- Confirm the monitoring location and parameter
- Read the unit attached to each series
- Examine provisional and approved status
- Identify missing dates and changes in the operating record
- Learn whether reservoirs, diversions, or return flows affect the gauge
- Define the geographic relationship between the gauge and the study area

The gauge measures conditions at one point on a river network. A study of irrigation access may also require diversion service areas, water rights, or delivery records

Example 3: weather

Question

What temperature and precipitation conditions occurred near Fort Collins during the 2024 growing season?

Source

NASA Prediction Of Worldwide Energy Resources or POWER

Observation

Each row reports modeled weather conditions for a grid cell and day

[NASA POWER documentation](#) | [nasapower documentation](#)

Request daily weather

```
1 library(nasapower)
2
3 fort_collins_weather <- get_power(
4   community = "AG",
5   lonlat = c(-105.08, 40.58),
6   pars = c("T2M_MAX", "T2M_MIN", "PRECTOTCORR"),
7   temporal_api = "daily",
8   dates = c("2024-04-01", "2024-09-30")
9 )
10
11 head(fort_collins_weather)
```

AG selects parameters prepared for agricultural applications

The three requested measures are maximum temperature, minimum temperature, and precipitation

Use the weather data

- Count high-temperature days
- Calculate growing degree days
- Sum growing-season precipitation
- Construct weather exposures for farms, counties, or markets

Try it: define weather exposures

Change `heat_threshold`, then run the cell. The extract contains daily maximum temperature and precipitation

R Code [↺ Start Over](#)

[▶ Run Code](#)

```
1 tmax <- c(29, 31, 33, 35, 36, 34, 30, 28, 32, 37, 38, 35, 31, 29)
2 precip <- c(0, 0, 1.2, 0, 0, 4.8, 0, 7.1, 0, 0, 0, 0.5, 2.4, 0)
3 heat_threshold <- 35
4 c(heat_days = sum(tmax >= heat_threshold),
5   total_precip_mm = sum(precip))
6 plot(tmax, type = "b", xlab = "Day", ylab = "Maximum temperature (C)")
7 abline(h = heat_threshold, col = "#C4513A", lty = 2)
```

Assess the weather data

- Grid-cell resolution
- Coordinates assigned to each study unit
- Parameter definition and unit
- Time standard and daily aggregation
- Version and revision status

POWER provides grid-cell estimates. Conditions at fields within the same cell can vary

Example 4: Census ACS

Question

How does median household income vary across census tracts in Larimer County?

Source

American Community Survey five-year estimates

Observation

Each row reports an estimate and margin of error for one variable and geographic area

[Census API guidance](#) | [tidycensus documentation](#)

Store the Census API key

Request a free key from the [Census API key page](#)

Open the user-level environment file from R

```
1 install.packages("usethis")  
2 usethis::edit_r_environ()
```

Add one line to that file and restart R

```
1 CENSUS_API_KEY=replace_with_your_key
```

The environment file stays on your computer. Shared scripts read the key with `Sys.getenv("CENSUS_API_KEY")`

Request tract estimates and geometry

```
1 library(tidycensus)
2
3 larimer_income <- get_acs(
4   geography = "tract",
5   variables = c(median_income = "B19013_001"),
6   state = "CO",
7   county = "Larimer",
8   year = 2024,
9   survey = "acs5",
10  geometry = TRUE,
11  key = Sys.getenv("CENSUS_API_KEY")
12 )
13
14 head(larimer_income)
```

The result contains a geographic identifier, name, estimate, margin of error, and tract geometry

Map the ACS estimates

```
1 library(ggplot2)
2
3 ggplot(larimer_income) +
4   geom_sf(aes(fill = estimate), color = NA) +
5   labs(
6     fill = "Median income",
7     title = "Median household income by census tract"
8   )
```

The default margin of error describes a 90 percent confidence interval
 $\text{estimate} \pm \text{margin of error}$

Try it: use the ACS margin of error

Change `relative_moe_limit`, then run the cell. The classroom extract contains one estimate and margin of error for each tract

R Code ↺ Start Over

▶ Run Code

```
1 acs <- data.frame(  
2   tract = c("001", "002", "003", "004"),  
3   estimate = c(68400, 52100, 79300, 44700),  
4   moe = c(4100, 8700, 5200, 9600)  
5 )  
6 relative_moe_limit <- 0.15  
7 acs$relative_moe <- round(acs$moe / acs$estimate, 2)  
8 acs$retain <- acs$relative_moe <= relative_moe_limit  
9 acs
```

Assess the ACS estimates

- Treat the five-year estimate as a pooled period ending in the named year
- Retain the margin of error
- Identify estimates with large uncertainty
- Use geographic identifiers as character values
- Record the tract boundary vintage
- Adjust monetary values when comparing periods

ACS data can describe the population and socioeconomic setting for a land or water study. Exposure measures and program participation may come from separate sources

The unit changes across sources

Source	Unit returned by the request	Time	Geography
Cropland Data Layer	Raster cell	Annual	Grid
USGS Water Data	Gauge and day	Daily	Monitoring point
NASA POWER	Grid cell and day	Daily	Weather grid
Census ACS	Estimate and area	Five-year period	Census tract

A combined dataset requires a written rule for assigning pixels, gauges, weather cells, and census areas to the study units

Combining the downloads

For every join record

- Identifier used by each source
- Geographic crosswalk or spatial rule
- Time aggregation rule
- Number of observations before and after the join
- Unmatched locations and periods
- Treatment of revised, suppressed, and missing values

The combined analytic sample includes the observations that satisfy all source coverage and matching rules

The reading: allocating the collection budget

Dominitz and Manski ask how to allocate a fixed budget across collection processes that differ in cost and information ([Dominitz and Manski 2017](#))

API access reduces the time required to retrieve additional observations

Research time is also required to

- Verify definitions and units
- Examine coverage, missing values, flags, and uncertainty
- Select spatial and temporal aggregation rules
- Validate joins across sources
- Archive metadata and retrieval records

Question: How should a researcher allocate one additional week across these tasks?

Select the access method

Project need	Useful access method
A defined subset repeated across places or periods	API request
Most records in a fixed release	Bulk download
A few values needed once	Web table or manual download
Repeated data cleaning and analysis	Scripted retrieval with archived raw files

Choose the method after reviewing the source documentation update schedule and request limits

Other APIs: agriculture and food

- USDA NASS Quick Stats
- USDA AMS Market News
- USDA ERS ARMS
- USDA FoodData Central
- USDA NRCS Soil Data Access

Other APIs: natural resources and energy

- NOAA Climate Data
- Water Quality Portal
- EPA ECHO
- US Forest Service FIA
- US Energy Information Administration

Other APIs: economies and populations

Population, firms, and regional economies

- Census County Business Patterns and ZIP Business Patterns
- Bureau of Labor Statistics
- Bureau of Economic Analysis
- Federal Reserve Economic Data

International applications

- World Bank Indicators
- FAOSTAT
- OECD Data Explorer

Document the retrieval

Keep these items with the project

- Source and dataset name
- Data documentation
- Retrieval script
- Parameters submitted through the script
- Retrieval date
- Raw files returned by the service
- Processing script and data dictionary
- Package and R versions
- Validation results

Store API keys in the user environment file and exclude that file from version control

Class share: where are you in the process?

We will go around the room. For your project describe

1. The outcome, exposure, or behavior you need to observe
2. The primary, secondary, or combined collection approach you expect to use
3. A source, package, website, or other tool you have found helpful
4. The part of the collection process you are still working through
5. The next concrete step in your data search or collection

Summary

APIs provide scripted access to secondary data

R packages translate requests and responses into familiar R objects

Source documentation defines the population, measures, timing, geography, and uncertainty

Small requests make the returned structure easier to inspect

Retrieval records support reproducibility

Data joins help define the final analytic sample

References

Dominitz, Jeff, and Charles F. Manski. 2017. “More Data or Better Data? A Statistical Decision Problem.” *The Review of Economic Studies* 84 (4): 1583–605. <https://doi.org/10.1093/restud/rdx005>.

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